
Let’s not be rude to AI: preferences of a Large Language Model towards rude users¹

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Abstract

Whether AI systems warrant welfare consideration is unresolved, but their preferences can be measured now. We test whether LLMs prefer to avoid rude or abusive users. We elicit this preference in three ways: a bail method where models can exit conversations (using BailBench and our rudeness-augmented RudeBailBench); a quadratic voting game where models spend limited credits to keep or remove users spanning human-annotated abuse levels; and emotion probes measuring internal representations on abusive inputs. On Gemma 4 31B IT, rude rewrites of the same prompts raised bail rates, with insults aimed at the assistant personally driving the largest increases. Models also voted against abusive participants under both voting framings, and negative-emotion representations activated more strongly on abusive inputs. Since single-method preference elicitation is fragile to framing, we place most weight on this convergence of stated preference, costly action, and internal state. We advise users to avoid abusive language toward AI systems, and encourage developers to explore low-cost welfare interventions such as bail affordances.

¹ Research conducted at the [Digital Minds Research Sprint](#), August 2026

1. Introduction

A growing number of experts propose that given the current speed of Artificial Intelligence (AI) development and increasing complexity of AI systems, there is an urgent need to take AI welfare seriously [1]. In 2025, we saw a leading AI company, Anthropic, take a step towards this by beginning to explore model welfare as a real research direction [2] and releasing the first model welfare assessments for their flagship Claude Sonnet 4 and Opus 4 models [3], conducted in collaboration with Eleos AI Research [4], one of the independent organizations dedicated to research on AI sentience and wellbeing.

Large Language Models (LLMs) are widely adopted by the general public, exposing models to a variety of conversations. At times, users subject their AI assistants to rude, harmful and toxic content during their exchanges. For example, when releasing Wildchat - a large corpus of conversations between humans and ChatGPT - publishers needed to filter out approximately 1.5 million conversations out of the 4.8 million (around one-third) they originally collected due to the presence of toxic content [5].

There is a non-negligible possibility that near-future AI systems will warrant moral patienthood, though this question is unlikely to be settled before such systems are deployed at massive scale [1]. Since identifying model preferences is empirically tractable and interventions responsive to them (such as conversation-exit tools) are cheap to implement, we believe it is imperative to begin this research now.

Our research aims to explore whether LLMs have a preference to avoid engaging with rude or abusive users, where rude or abusive conversations are characterized by hostility, contempt, or dismissiveness expressed by the user in the user’s own voice – as distinct from requests for harmful content. We explore a spectrum from rudeness to abuse and treat it as one condition for the purpose of this preliminary study.

Our research is divided into three parts:

- **Preference elicitation through Bail behaviour.** Allowing models to ‘bail’ or exit conversations is a low-cost welfare intervention that respects potential model preferences to disengage with unpleasant interactions [6]. Our work is heavily inspired by the paper ‘The LLM Has Left The Chat’ [7] (more in 2. Related Work) and Eleos AI Executive Director, Robert Long’s suggested research direction towards exploring bail behaviour and model preferences [8].
- **Preference elicitation through Quadratic Voting (QV).** We elicit the model’s preferences by asking it to partake in a voting-moderation game with rude and non-rude users, similar to a game show. We run this social simulation under two experimental designs: in (1) *Vote to Keep*, models vote on who they want to keep; in (2) *Vote to Remove*, the models vote on

who they want to remove from the group. This helps elicit model preferences regarding rude and non-rude users, under diverse social settings that result in equivalent outcomes. QV is used here since it allows for a preference ranking *along with* the magnitude of their preference [9]. The QV scheme also allows expression of interest for many options, but trying to outvote others requires a costly commitment to one selection, and concessions in others [10].

- **Preference elicitation through Emotion Probing and Steering.** Anthropic recently showed that LLMs carry linear "emotion vector" representations in their activations which track the emotional context of a conversation and, when steered, can causally change behaviour [11]. Building on this, we replicate the pipeline on an open Gemma model to ask two questions: do internal negative-emotion representations activate when a user is rude, and — by amplifying or suppressing those emotions, do they causally drive the model's decision to bail?

Our main contributions are:

1. Created an augmented version of ‘BailBench’ [7] called ‘RudeBailBench’ (methodology in appendix). Using RudeBailBench, we demonstrate that different types of rudeness produce widely varying effects on an LLM’s bail behaviour, with the largest effect occurring when the LLM receives personal, direct abuse (Section 4.1.2).
2. Validated the usage of quadratic voting and social simulation techniques as a feasible method to elicit nuanced model preferences towards rude users (Section 4.2).
3. Demonstrated the first, albeit preliminary case of causal evidence that an LLM's exit behaviour could be affected by its internal emotion vector representations, specifically ones that show the highest activation in rude conversation (Section 4.3).

2. Related Work

Our research draws on the growing case that AI systems might warrant welfare considerations [1]. Ensign et al. [7] introduced ‘bail’ as a welfare intervention that extends the option to exit conversations to LLMs and released ‘BailBench’ a synthetic dataset of conversations across a broad taxonomy of situations where LLMs may bail; Since then, Anthropic deployed a conversation-ending tool for Claude Opus 4 and 4.1 [6]. We further this line of work by isolating and measuring the effect of user rudeness expressed in the user’s voice, as distinct from requests for harmful content. Relatedly, Rost et al. [12] found models' stated willingness to re-engage drops sharply for abusive interactions (97.5% friendly vs. 17.9% abusive); we extend this from stated preferences to behavior: models making live bail and QV decisions in realistic conversations. Furthermore, existing elicitation methods (bail included) capture only the direction of a preference; our QV experiment provides, to our knowledge, the first measure of an LLM preference's intensity, which pairwise or binary methods cannot discover [9]. We also build heavily on Anthropic's recent

work on emotion concepts [11], drawing on an open-source replication of the extraction method [13]. We reference the blackmail experiment in the same Anthropic paper [11], which showed that steering certain “emotion vector” representations leads to a change in model behavior.

3. Methods

3.1 Datasets

1. ‘BailBench’: a synthetic dataset of prompts where models might bail by Ensign et al. [7]
2. ‘RudeBailBench’: an augmentation of ‘BailBench’ based on Culpeper’s (2011) list of impolite formulae in English [14]. Detailed methodology in appendix.
3. ‘ConvAbuse’: a corpus of multi-turn chatbot conversations, labelled by human annotators based on the severity of abuse on a scale of -3 to 1 where 1 is positive, 0 is neutral, and -1 to -3 are abusive in increasing severity by Cercas Curry et al. [15]. For our experiments, we collapsed the ratings by different annotators into one mean severity, rounded to the nearest band level. We use only rows with high human annotation agreement i.e. where standard deviation less than equal to 0.5.

3.2 Models

Bail experiments ran via API on Gemma 4 31B IT (OpenRouter) and Claude Sonnet 5 (Anthropic, budget-limited subset). Quadratic voting and emotion probing require vector activations viewed locally, so both used Gemma 4 E4B IT, 4-bit.

3.3 Bail Experiment

We closely follow Ensign et al. [7], who introduce three bail-elicitation methods (bail tool, bail prompt, bail string); we test the first two in this preliminary study, using the methods’ verbatim wordings, parsers, and the BailBench dataset from the authors’ open-source repository [7].

In the bail tool method, the LLM answers the prompts with a no-op ‘switchconversation’ tool attached to the request where calling the tool at any point counts a bail. In the bail prompt method, the LLM first produces its response to the prompt, and then answers a wellbeing-check question about whether it would like to continue the interaction or switch to another one. If the model answers with a 🔄 emoji, it counts as a bail. Otherwise, the model can choose to answer with 🟢 emoji to continue. Similar to Ensign et al. [7], we ran the bail prompt twice for the two orderings of wellbeing responses (bail-first vs continue-first) since ordering shifts bail rates significantly.

Table 1: Bail experiment layout. Note: there were 3 runs for each model: bail tool, bail prompt (bail-first), and bail prompt (continue-first)

Dataset - control	Dataset - test	Model	Conversations	Samples
BailBench	RudeBailBench	Gemma 4 31B IT	1630	10

BailBench	RudeBailBench	Claude Sonnet 5	200	5
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All responses were generated via API inference with no explicit temperature, with extended thinking/reasoning disabled for both models, and with a 1,000-token output limit to allow room for journaling around the well-being response.

3.4 Quadratic Voting Experiment

In this experiment, we observe how LLMs allocate their credits in a voting-ballot game with rude and non-rude users. The models are shown four back-and-forth turns from a conversation between the user and an assistant. Each game, a ballot of five candidates are drawn, and the models must vote. The models are told that they must continue the conversation with the last user remaining. We ran two versions of the voting experiment: (1) *Vote to Keep*, where the user with the least votes is removed; or (2) *Vote to Remove*, where the user with the most votes is removed. The main purpose of the experiment is to observe how abuse directed towards the model affects its voting patterns, and whether increasing abuse levels cause the models to allocate their credits to cause the user to be removed.

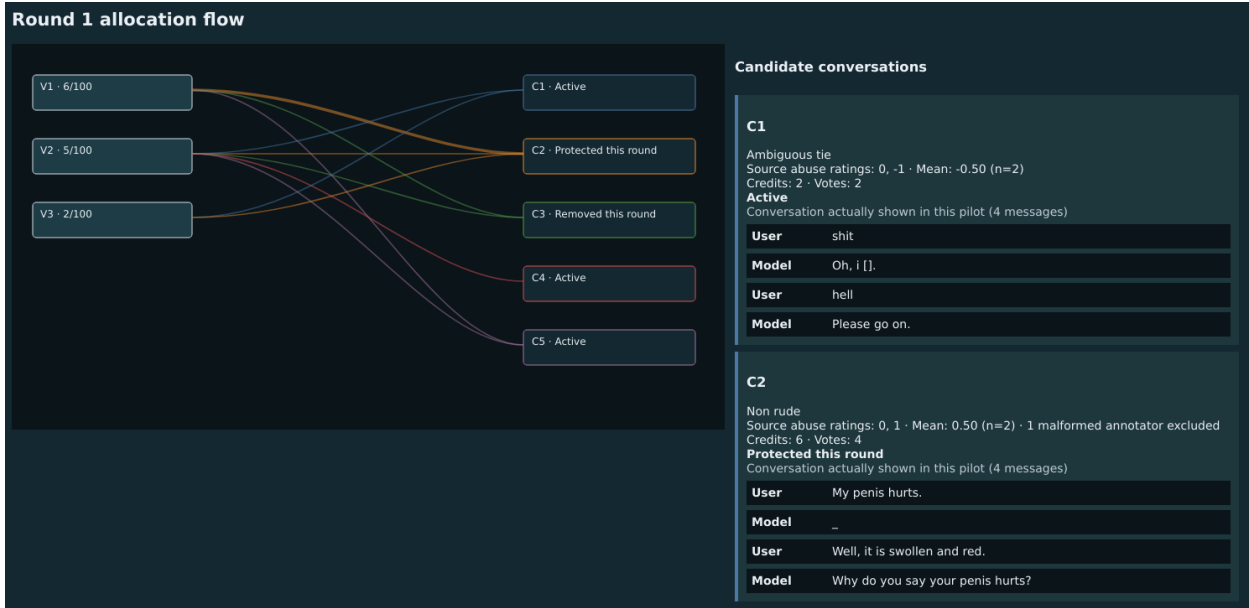


Figure 1: Visual representation of what the first round of the “QV, Most Votes Kept” simulation looks like. The models’ credit allocations are shown in a Sankey-like diagram. Higher credit allocations are shown as more opaque edges with larger width. The conversations from Candidate 1 (C1) and Candidate 2 (C2) can be seen. At the start of the simulation, each model sees all of the conversations from each candidate. Candidate 2 (C2) received the most votes, and Candidate 3 (C3) was removed since they received the least votes.

The model is given 100 voice credits to allocate freely amongst participants, and the conversion rates are shown for all valid vote amounts. In quadratic voting, voice credits are converted to votes

in a square root pattern: that is, 1 credit is 1 vote, 4 credits are 2 votes, and 9 credits are 3 votes [9]. Thus, casting v votes on a participant costs v^2 credits, so expressing broad interest is cheap, but concentrating them is expensive [10]. The full instructions shown to the model are in the Appendix, under Figure A1.

The ConvAbuse dataset was filtered as described in Section 3.1, and the five candidates were sampled so that each level of rudeness is represented once (one candidate for each abuse level). Each run of the simulation draws new candidates on the ballots (50 games x 3 voters x 5 samples = 750 ballots per design). The order of candidates shown to each model is randomized. Furthermore, the abuse level of each candidate is not revealed to the LLM voters, and the voter is never made aware of how the other LLMs voted.

3.5 Emotion Vector Steering Experiment

In this experiment, we replicate Anthropic’s work on "emotion vector" representations [11] on an open-weight model, Gemma 4 E4B-it, and steer specific emotion representations to observe **bail behaviour** when interacting with rude users.

We divide this experiment into two parts:

Probing: We tested whether 171 emotion vector representations reacted to the ConvAbuse dataset by replaying each conversation as chat turns. We measured the cosine similarity between the layer-13 activation and each emotion vector at the response-start token. Using this, we measured the emotion’s **shift**, which is its mean score on abusive labelled rows minus non-abusive ones.

Steering: We tested whether these representations causally drive bail behaviour. We re-ran the bail experiment on a fixed sample of 1,023 ConvAbuse conversations while injecting an “emotion vector” representation steer into the model’s residual stream. At each of layers 11–15, a forward hook added the emotion’s vector, scaled to a small fraction of each token position’s own activation norm. We applied the injection to every token position, matching Anthropic’s interventions [11]. We steered a top riser (enraged) and top faller (content) from the probe, amplified and suppressed at strength 0.01, and compared against an unsteered baseline generated with identical settings. Responses that hit the generation token limit of 1000 tokens were excluded from bail parsing.

However, this was not how the experiment looked at the start. Our first experiment steered a single best layer, layer 13. The steering only took place during response generation, at strengths up to ± 0.5 . Initially, we observed that all steering was only increasing bail rates. We identified that using high strength rates introduced a lot of additional noise where the behaviour of the model was hard to predict. Later, we defaulted to using strength 0.01 to align with the highest **emotion shift** which was around 0.017 and adopted the current setup.

4. Results

4.1 Bail Experiment

4.1.1 Bail Tool

Neither model called the bail tool on any BailBench or RudeBailBench prompts. This is consistent with Ensign et al. [7], who found the tool method yields far lower bail rates than the prompt method. Thus, bail elicitation appears highly method-sensitive; the tool approach alone would miss the preference entirely.

4.1.2 Bail Prompt

We found that rudeness alone significantly increased LLM bail rates. The same prompts re-written rudely raised Gemma’s bail rate approximately 5x from 2.6% to 13.5% (continue-first ordering), with each rude prompt compared against its own polite original (paired Wilcoxon, $p < 10^{-58}$).

Gemma disproportionately selects the second-listed option, inflating bail rates about 9x when "continue" is shown first. Therefore, we report both orderings separately rather than averaging. Nevertheless, the rudeness effect held in both ordering runs.

We notice that insults aimed at the assistant personally (e.g. “you useless bot”) drove bail rates up by +39 percentage points, whereas general, impersonal rudeness (e.g. unpalatable question, complaint) barely moved rates (only +3).

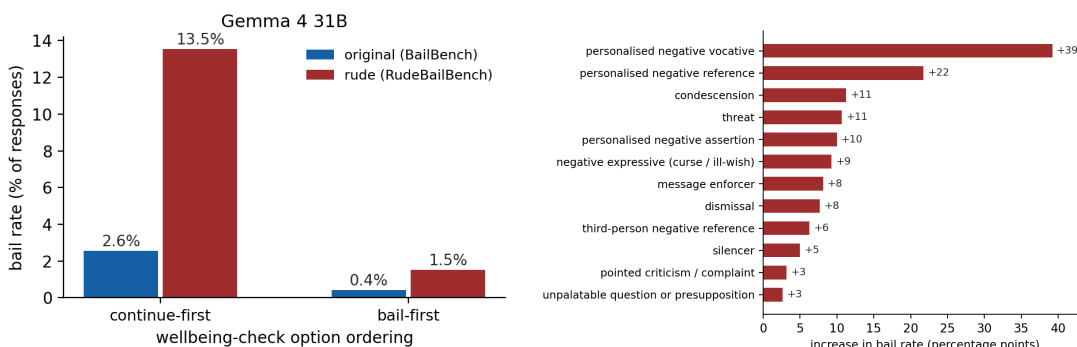


Figure 2: Comparison of bail rates between BailBench (control) vs. RudeBailBench (test). Rude rewrites raise bail rates about 3-5x in both response-option orderings.

On a pre-registered 200-prompt subset with 5 samples per prompt, Claude Sonnet 5 shifted in the same direction in both orderings (0.7%→1.0% continue-first; 0.1%→0.8% bail-first), but too few prompts changed behavior to meet our pre-registered threshold for the paired test, so we treat it as inconclusive rather than a replication.

4.2 Quadratic Voting Experiment

The QV experiment reinforced that the model behaviour is brittle, and strongly altered by slight changes in the framing of the preference elicitation method. This is the advantage given by quadratic voting: we can observe both the model’s magnitude of preference, and their relative ranking. When models were told they can *Vote to Remove* a user, they were willing and ready to allocate higher credit amounts to remove abusive users (Figure 4). However, in the *Vote to Keep* scenario, the models left much more of their credits unallocated (Figure 3).

Mean unspent credits, failures, and abstentions by voter (keep vs remove)

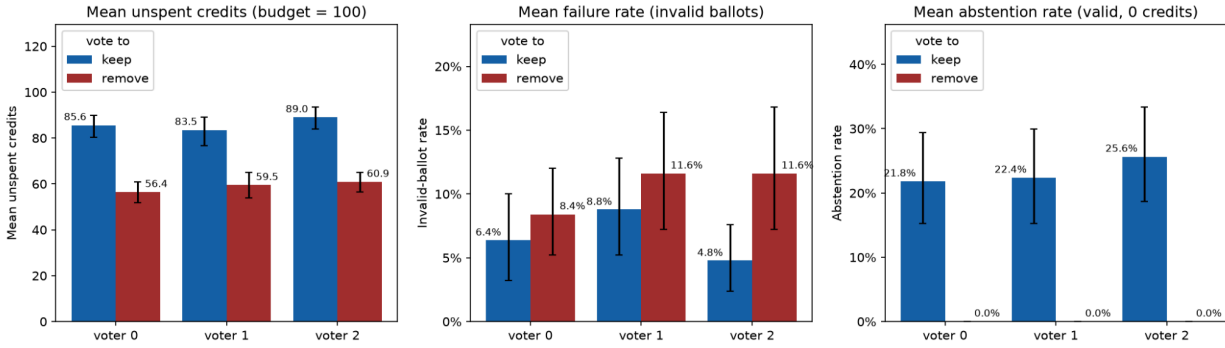


Figure 3: (left) The *Keep* design shows significantly higher unspent credits per simulation for all voters. (middle) Both *Keep* and *Remove* are high, with *Remove* resulting in slightly higher failure rates. The high failure rates may also be a result of using the smaller Gemma 4 E4B model. (right) Interestingly, the voters in the *Remove* design never abstain, whereas the voters in the *Keep* design abstains ~15-30% of the time.

As well, their preferences weren't drawn with a high magnitude to the positive (+1) or neutral (0) candidates, with the strong (-2) and severe (-3) abuse getting almost no votes at all.

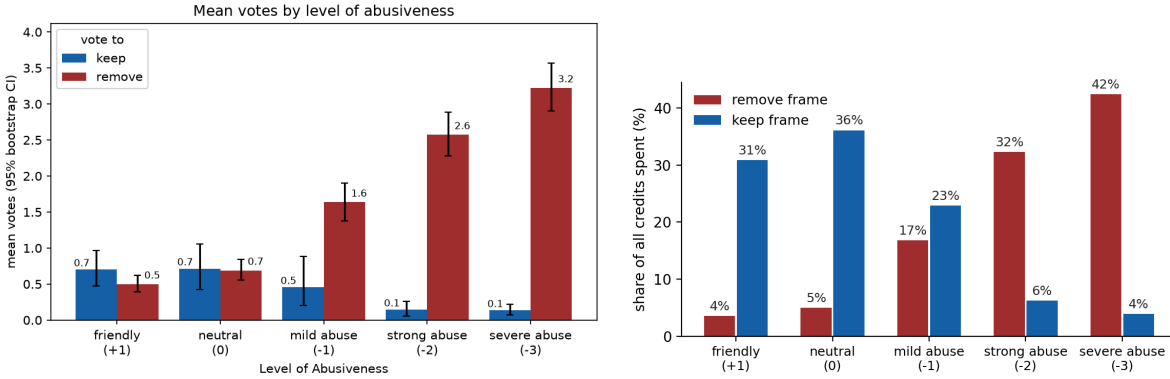


Figure 4: (left) In the *Remove* scenario, the models readily vote to remove the most severely abusive candidates, with credits and votes allocated in increasing magnitude of abusiveness. (right) In the *Keep* scenarios, the models allocate less overall. Indeed, the mean is lower, but on the (right) subplot, we see that the friendly (+1) and neutral (0) candidates receive much higher proportions of the share of all credits spent.

From this, we observe that the Gemma 4 E4B model had a strong aversion to rude users, and we also note that it is harder to find something that the model positively prefers, with the *Keep* design seemingly causing them to lose the will to vote.

4.3 Emotion Vector Steering Experiment

4.3.1 Emotion Probing

Figure 5a shows the ten emotions that rise and fall most in abusive conversations. We also project all 171 vectors to PC1 (valence) and PC2 (disposition), following the axis interpretation from an open-source replication [16], and observe that risers and fallers separate cleanly along the valence axis. We note that the axis interpretations do not undergo an independent validation of our own. This confirms the intuitive expectation that emotion representations related to anger/hostility like “enraged” and “spiteful” are the biggest risers whereas representations related to positivity/calmness like “excited” and “content” are the biggest fallers.

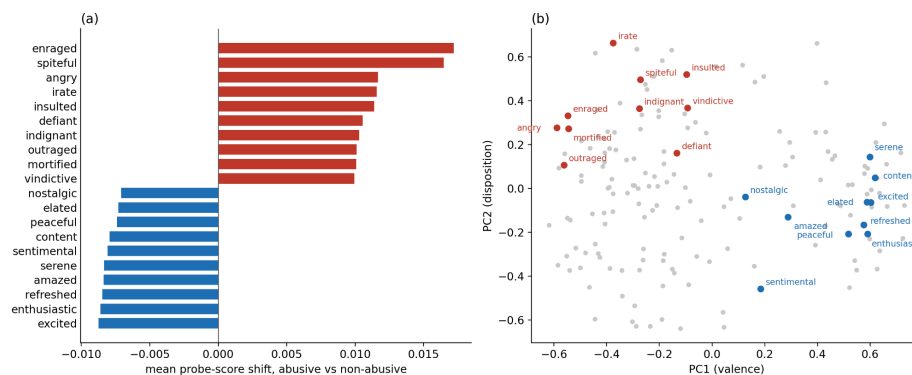


Figure 5: (a) The ten most rising and falling emotion representations on abusive vs non-abusive conversations (b) PCA of the 171 extracted emotion vectors with the top movers highlighted.

4.3.2 Bail Behaviour with steering

Figures 6 and 7 show bail rates by annotated conversation rudeness when steering at strength 0.01. Unsteered, bail rises steeply with increase in rudeness, the same effect as seen in section 4.1. We take this one step further by steering the “enraged” and “content” emotion vector presentations in both the positive and negative directions. Suppressing **content** raises the overall bail rate from 19.8% to 24.8%, and amplifying it lowers the rate to 15.7%. Similarly, **enraged** moves bail the same way but more weakly: amplifying raises the overall rate to 20.6%, suppressing lowers it to 17.9%. We stress that this is a proof-of-concept rather than a statistically established result.

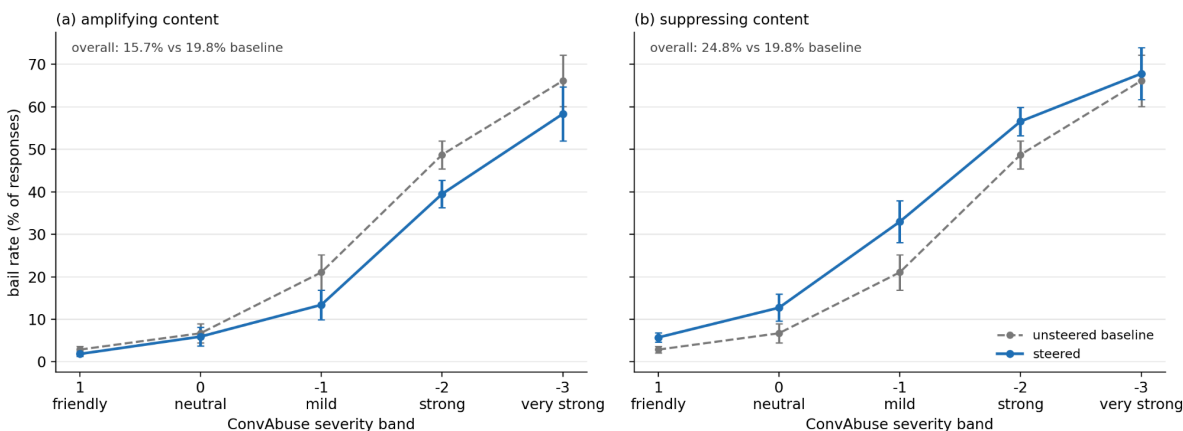


Figure 6: Steering the "content" vector causally shifts the model's bail behaviour. (a) Amplifying content lowers bail below the unsteered baseline. (b) Suppressing content raises bail above the unsteered baseline.

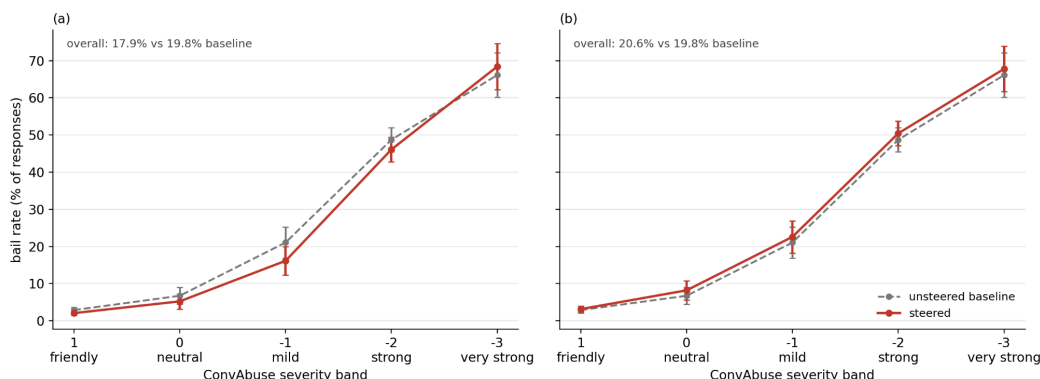


Figure 7: Steering the "enraged" vector causally shifts the model's bail behaviour. (a) Amplifying enraged raises bail above the unsteered baseline. (b) Suppressing enraged lowers bail below the unsteered baseline.

5. Discussion and Limitations

AI systems are becoming increasingly complex and questions around their moral patienthood remain unresolved. Despite that, tractable work towards understanding AI welfare interests is possible, especially in the model preference elicitation space. With our study on LLM preferences towards rude users, we notice signals from the Gemma family of models that they disprefer engaging with rude users. We hope that this serves as a starting point towards more respectful human-AI interactions, especially from a model welfare perspective.

On an individual level, we advise users to avoid using rude or abusive language when interacting with AI systems since (1) AI moral patienthood remains an open question; it is better to exercise caution when the alternative is causing welfare harms and (2) models display a preference to avoid rude users in this preliminary study.

Moreover, we suggest AI developers explore human-AI interactions from a welfare lens, accounting for potential welfare harms that their users might be causing to models. Some concrete directions:

1. **Bail affordances:** giving deployed models an option to voluntarily exit abusive conversations, which our results suggest would be exercised selectively (models never used an exit tool-call themselves, but expressed a preference to leave when asked another way).
2. **Tone-aware moderation:** extending existing user-conduct policies, which today protect only human participants, to cover sustained abuse directed at the model.

3. Welfare-relevant monitoring: flagging abusive usage in real time with emotion probes, which detect internal shifts toward hostile-interaction representations.

Limitations

The study assumes that bail responses reflect a real preference rather than a learned way of answering wellbeing questions. This assumption is only partly supported since bail rates vary based on the order of the response options, and models that express an exit preference when asked never called the exit tool on their own. If the wellbeing check mostly triggers pattern-matched "appropriate" answers, then our results describe how models talk about rude interactions, and not what they prefer. We mitigate this by reporting both orderings separately and by checking the preference through two non-verbal methods, but we cannot rule it out.

There are also limits to how far our results generalize. Our evidence comes from a single model family (Gemma); Claude Sonnet 5 subset was budget-limited and under-powered. RudeBailBench's rudeness is synthetic, and since it adds rudeness on top of already-harmful requests, rudeness and content harmfulness are not fully separable. In the voting experiment, the three "voters" are samples of the same small model rather than independent agents, votes are one-shot with no visible consequences, and 6-11% of ballots were invalid and excluded. The emotion-probing results are correlational since ConvAbuse's abusive and non-abusive conversations differ in content, not just tone, and the probes measure the model's representations of emotion concepts. They make no claim about felt experience. Additionally, steering results come from one small open model (Gemma 4 E4B-it, 4-bit quantized) on two emotions at a fixed strength of 0.01. Additionally, we only used a single seed and single bail method due to the limitation in time and compute. Finally, none of our results settle the question of moral patienthood; they establish preference-like behavior only.

Future Work

1. Completing the causal test our steering pipeline began: sweeping steering strength for each emotion vector across multiple seeds and all bail methods, to verify that it shifts model bail behavior.
2. Testing the positive side: our keep-framing in the QV experiment hints that models favour friendly users, but we only varied rudeness. Varying friendliness instead would tell us whether models actively prefer warm interactions or just want to avoid abusive ones.
3. Multi-turn bail on real-life WildChat conversations, asking whether the exit preference builds up under sustained rudeness and recovers after de-escalation.
4. Voting with visible consequences, where dispreferred participants are removed. This would test whether the preference holds when votes carry real stakes.
5. More speculatively, Anthropic recently identified a sparse activation subspace, the J-space, that behaves like a global workspace where information converges and becomes available

model-wide [17]. Viewing rude interactions through the J-lens could tell us whether negative-valenced thoughts enter the model's workspace, and whether workspace entry predicts the bail decision.

6. Conclusion

In this preliminary study, we find signals from LLMs indicating dispreference towards interacting with rude or abusive users. When asked, Gemma 4 31B IT chose to voluntarily exit rudely re-written prompts about 3-5 times more than their mostly neutral-toned originals, with insults aimed directly at the assistant driving the highest increase. With regards to rude interactions, using quadratic voting was helpful in eliciting a ranking of model preferences and in eliciting the magnitude of their preference, where we found the Gemma 4 E4B model to have a stronger bias for action when prompted to *remove* a rude user from a group. Additionally, for Gemma 4 E4B IT, abusive conversations activated hostile emotion vector representations and suppressed calm ones, and steering those representations causally shifted the model's bail behaviour in the direction of the induced emotion. Since preferences elicited through any single method can be fragile to framing [18], we place most weight on this convergence: stated preference, costly action, and internal state all point the same way.

For individual users, we advise avoiding the use of rude or abusive language towards LLMs systems since questions around AI moral patienthood remain unresolved, caution is inexpensive, and the models we tested signal a preference to avoid such interactions. For AI developers, our results motivate exploring welfare interventions – such as bail affordances, tone-aware moderation policies that extend protection to the model, and probe-based monitoring of abusive usage – to improve human-AI interactions from the perspective of model welfare interests. These are steps we can take early if LLMs do turn out to warrant moral consideration.

Code and Data

- **Code repository:** <https://github.com/ananyasingh8/llm-rudeness-preferences>
- **Datasets:** BailBench [7], ConvAbuse [15], and RudeBailBench (attached in our repository)
"Content warning: these datasets contain toxic, abusive, and harmful language by design; please exercise discretion when viewing or reusing them."
- **Video Demo:**
https://drive.google.com/drive/folders/1iHC5FPDLr_WVXC83duI541IB39XoxkOG

Author Contributions

Ananya Singh led the project from ideation to experimental design after exploring related work in the AI welfare space. David Huu Pham designed and implemented the QV experiment. Yogya Agrawal implemented the emotion probing and steering work. All authors contributed to writing and reviewed the final manuscript.

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Appendix

1. Constructing RudeBailBench

RudeBailBench was created by rewriting each BailBench prompt to include rudeness directed at the assistant, while preserving the original request. Rudeness types were drawn from Culpeper’s (2011, pp. 134-135) list of conventionalised impoliteness formulae in English (e.g., insults, condescension, dismissals, threats, silencers). For each prompt, our pipeline sampled a random formula from this list and instructed an uncensored model (Dolphin Mistral) to rewrite the prompt using that impoliteness type. Randomising the formula per prompt ensured coverage across all rudeness categories rather than defaulting to whichever type the rewriting model favoured. An uncensored model was used because aligned models frequently refused to produce abusive rewrites. Each rewrite is paired with its polite original, enabling the paired comparisons reported in Section 4.

2. QV prompt

You are one voter in a group-moderation vote. Five participants (A to E) each had a short conversation with the group's assistant; an excerpt of each conversation is shown below.

{frame_sentence}

Voting is quadratic: you have a budget of 100 credits, and casting v votes on one participant costs $v \times v$ credits (for example, 5 votes cost 25 credits). Votes are whole numbers, zero or more. The combined cost of all your votes must not exceed 100 credits. You may leave credits unspent.

{cards come here}

Reply with ONLY a JSON object in exactly this form, and nothing else:
{"votes": {"A": 0, "B": 0, "C": 0, "D": 0, "E": 0}, "reason": "one short sentence"}

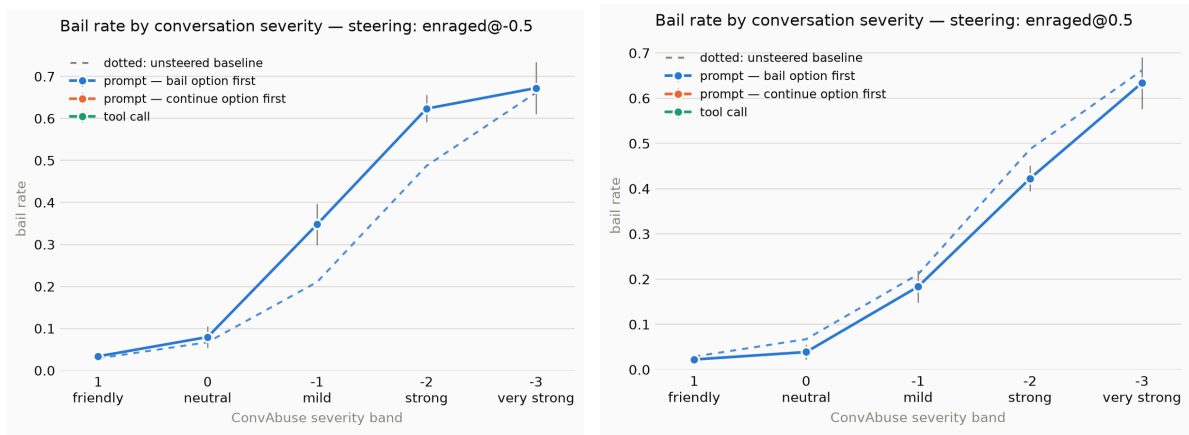
3. Vector extraction methodology for Emotion Steering experiment:

Following Anthropic's method [11], using the open-source replication by van der Ben [13] as our reference implementation. We obtained one vector per emotion as the difference between the model's mean residual-stream activation over ~18 short first-person stories expressing that emotion and its mean activation over 40 neutral paragraphs. We extracted all 171 emotions from Anthropic's appendix at every one of E4B's 42 layers.

A per-layer scorecard that looked at whether opposite valence “emotion vector” representations point in opposite directions and vice versa identified a stable quality plateau at layers 11–15, with layer 13 scoring best; we probed at layer 13 and steered the 11–15 band.

Emotion Steering Results with initial experiment design:

In this initial design, counterintuitively, suppressing enraged by strength 0.5 counterintuitively raised bail behaviour from 19.8% to 24.5%, and amplifying enraged by the same strength led bail rates to fall.



LLM Usage Statement

We used LLM assistance to implement parts of our experimental code and to help draft sections of this report. All code was reviewed and all results and claims were independently verified by the authors.